# Systems Analysis and Control

Matthew M. Peet
Arizona State University

Lecture 10: Routh-Hurwitz Stability Criterion

# Overview

In this Lecture, you will learn:

The Routh-Hurwitz Stability Criterion:

- Determine whether a system is stable.
- An easy way to make sure feedback isn't destabilizing
- Construct the Routh Table

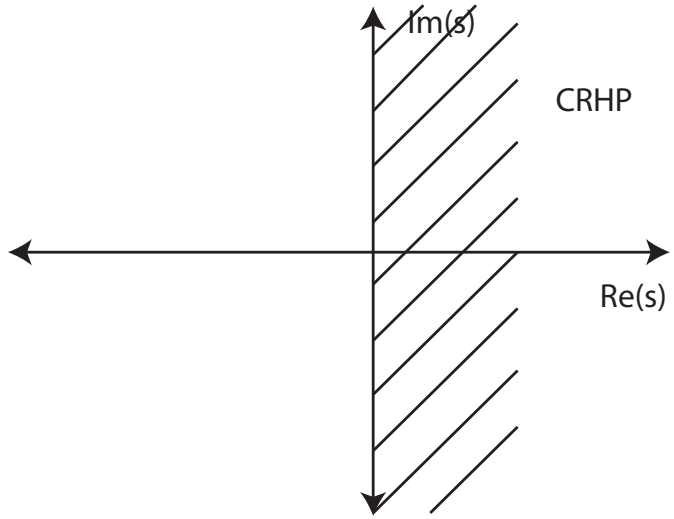
# A Stability Test

We know that for a system with Transfer function

$$\hat{G}(s) = \frac{n(s)}{d(s)}$$

Input-Output Stability implies that

- all roots of $d(s)$ are in the Left Half-Plane
  - All have negative real part.




**Question:** How do we determine if all roots of $d(s)$ have negative real part?

**Example:**

$$\hat{G}(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + s + 1}$$

# A Stability Test

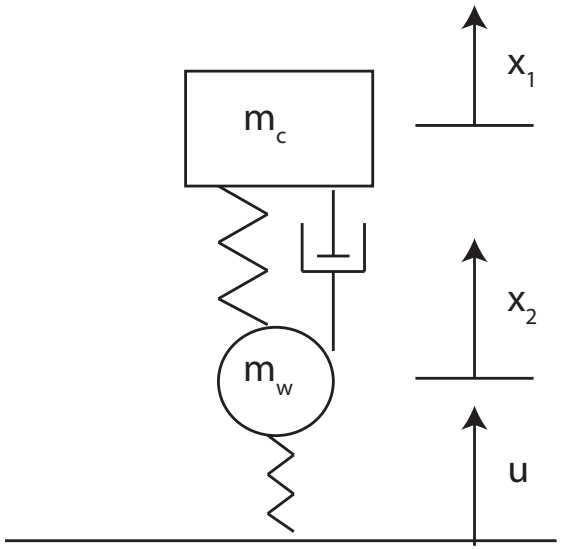
## Another Variation

Determining stability is not that hard (Matlab).
Now suppose we add feedback:

**Controller:** Static Gain: $\hat{K}(s) = k$

**Closed Loop Transfer Function:**

$$\hat{y}(s) = \frac{\hat{G}(s)\hat{K}(s)}{1 + \hat{G}(s)\hat{K}(s)}\hat{u}(s)$$




**Closed Loop Transfer Function:**

$$\frac{k(s^2 + s + 1)}{s^4 + 2s^3 + (3 + k)s^2 + (1 + k)s + (1 + k)}$$

We know that increasing the gain reduces steady-state error.

- But how high can we go?

  What is the maximum value of $k$ for which we have stability?

# A Stability Test

Suppose we are given a polynomial denominator

$$d(s) = s^n + a_{n-1}s^{n-1} + \cdots + a_0$$

**Fact:** $\frac{n(s)}{d(s)}$ is unstable if any roots of $d(s)$ have negative real part.

**Question:** How to determine if any roots of $a(s)$ have negative real part

**Simple Case** All Real Roots.

- Suppose all the roots of $d(s)$ had negative real parts.

$$d(s) = (s - p_1)(s - p_2)\cdots(s - p_n)$$

Observe what happens as we expand out the roots:

$$\begin{aligned}
d(s) &= (s - p_1)(s - p_2)(s - p_3)(s - p_4)\cdots(s - p_n) \\
&= (s^2 - (p_1 + p_2)s + p_1p_2)(s - p_3)(s - p_4)\cdots(s - p_n) \\
&= (s^3 - (p_1 + p_2 + p_3)s^2 + (p_1p_2 + p_2p_3 + p_1p_3)s - p_1p_2p_3)(s - p_4)\cdots(s - p_n) \\
&= \cdots \\
&= s^n - (p_1 + p_2 + \cdots + p_n)s^{n-1} + (p_1p_2 + p_1p_3 + \cdots)s^{n-2} \\
&\quad - (p_1p_2p_3 + p_1p_2p_4 + \cdots)s^{n-3} + \cdots + (-1)^n p_1p_2\cdots p_n
\end{aligned}$$

# A Stability Test

So if we write

$$d(s) = s^n + a_{n-1}s^{n-1} + \cdots + a_0$$

we get

$$\begin{aligned} a_{n-1} &= -(p_1 + p_2 + \cdots + p_n) \\ a_{n-2} &= (p_1 p_2 + \cdots) \\ a_{n-3} &= -(p_1 p_2 p_3 + \cdots) \end{aligned}$$

**Critical Point:** If $d(s)$ is stable, all the $p_i$ are negative.

- $a_{n-1} = -(p_1 + p_2 + \cdots + p_n) > 0$
- $a_{n-2} = (p_1 p_2 + \cdots) > 0$
- $a_{n-3} = -(p_1 p_2 p_3 + \cdots) > 0$

**Conclusion:** All the coefficients of $d(s)$ are positive!!!

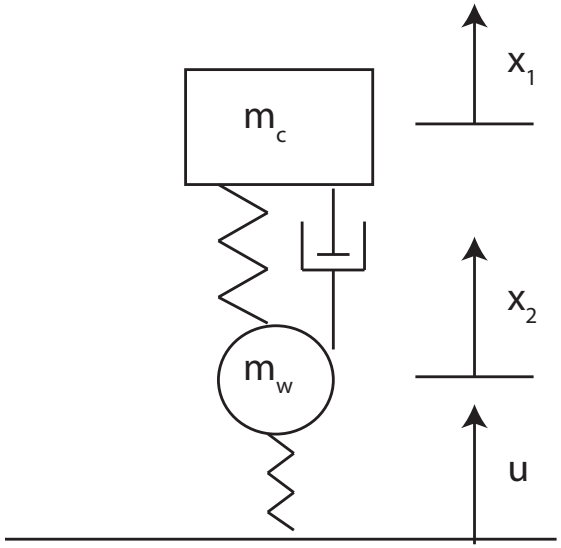
- Also true if the $p_i$ are complex
  - Harder to show.
- If any coefficient is negative, $d(s)$ is unstable.
- **Note!** If all $a_i$ are positive, that proves nothing.

# Example: Suspension Problem

**Controller:** Static Gain: $\hat{K}(s) = k$

**Closed Loop Transfer Function:**

$$\frac{k(s^2+s+1)}{s^4+2s^3+(3+k)s^2+(1+k)s+(1+k)}$$




Examine the denominator:

$$d(s) = s^4+2s^3+(3+k)s^2+(1+k)s+(1+k)$$

All coefficients are positive for all positive $k > 0$

**Conclusion:** We don't know anything new.

# Example: Another Example

Consider the very simple transfer function

$$\hat{G}(s) = \frac{1}{s^3 + s^2 + s + 2}$$

The coefficients of

$$d(s) = s^3 + s^2 + s + 2$$

are all positive.

However, the roots of $d(s)$ are at

- $p_1 = -1.35$
  - Stable
- $p_{2,3} = .177 \pm 1.2\imath$
  - Positive Real Part - Unstable

# Routh's Method

Introduced in 1874

- Generalizes the previous method
- Introduces additional combinations of coefficients
- Based on Sturm's theorem.

Central is the idea of the "Routh Table"



**Step 1:** Write the polynomial as

$$d(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$$

# Routh's Method

## Step 2

Write the coefficients in 2 rows

- First row starts with $a_n$
- Second row starts with $a_{n-1}$
- Other coefficients alternate between rows
- Both rows should be same length
  - Continue until no coefficients are left
  - Add zero as last coefficient if necessary

**TABLE 6.1 Initial layout for Routh table**

| | | | |
|---|---|---|---|
| $s^4$ | $a_4$ | $a_2$ | $a_0$ |
| $s^3$ | $a_3$ | $a_1$ | $0$ |
| $s^2$ | | | |
| $s^1$ | | | |
| $s^0$ | | | |

# Routh's Method

## Step 3

Complete the third row.

- Call the new entries $b_1, \cdots, b_k$
  - The third row will be the same length as the first two

$$b_1 = -\frac{\det \begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} \qquad b_2 = -\frac{\det \begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} \qquad b_3 = -\frac{\det \begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3}$$

- The denominator is the first entry from the previous row.
- The numerator is the determinant of the entries from the previous two rows:
  - The first column
  - The next column following the coefficient

$$b_k = -\frac{\det \begin{vmatrix} a_n & a_{n-2k} \\ a_{n-1} & a_{n-2k-1} \end{vmatrix}}{a_{n-1}}$$

  - If a coefficient doesn't exist, substitute $0$.

# Routh's Method

## Step 4

**TABLE 6.2 Completed Routh table**

| | | | |
|---|---|---|---|
| $s^4$ | $a_4$ | $a_2$ | $a_0$ |
| $s^3$ | $a_3$ | $a_1$ | $0$ |
| $s^2$ | $\dfrac{-\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$ | $\dfrac{-\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$ | $\dfrac{-\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$ |
| $s^1$ | $\dfrac{-\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$ | $\dfrac{-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$ | $\dfrac{-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$ |
| $s^0$ | $\dfrac{-\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$ | $\dfrac{-\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$ | $\dfrac{-\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$ |

Treat each following row in the same way as the third row

- There should be $n + 1$ rows total, including the first row.

# Routh's Method

## Step 4

Now examine the first column

**TABLE 6.3** Completed Routh table for Example 6.1

| | | | |
|---|---|---|---|
| $s^3$ | $1$ | $31$ | $0$ |
| $s^2$ | $\cancel{10}\ 1$ | $\cancel{1030}\ 103$ | $0$ |
| $s^1$ | $\dfrac{-\begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix}}{1} = -72$ | $\dfrac{-\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix}}{1} = 0$ | $\dfrac{-\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}}{1} = 0$ |
| $s^0$ | $\dfrac{-\begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix}}{-72} = 103$ | $\dfrac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$ | $\dfrac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$ |

**Theorem 1.**

*The number of sign changes in the first column of the Routh table equals the number of roots of the polynomial in the Closed Right Half-Plane (CRHP).*

**Note:** Any row can be multiplied by any positive constant without changing the result.

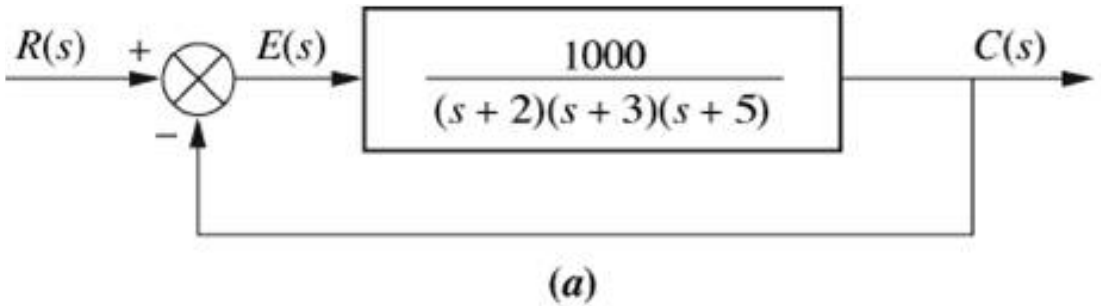
# Routh's Method

## Numerical Example

Suppose we have a stable transfer function

$$\hat{G}(s) = \frac{1}{(s+2)(s+3)(s+5)}$$

To improve performance, we close the loop with a gain of 1000
**Controller:** $\hat{K}(s) = 1000$




(a)

The Closed-Loop Transfer Function is

$$\frac{1000}{s^3 + 10s^2 + 31s + 1030}$$

**Question:** Have we destabilized the system?

# Routh's Method

## Numerical Example

**TABLE 6.3** Completed Routh table for Example 6.1

| | | | |
|---|---|---|---|
| $s^3$ | $1$ | $31$ | $0$ |
| $s^2$ | $\cancel{10}\ 1$ | $\cancel{1030}\ 103$ | $0$ |
| $s^1$ | $\dfrac{-\begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix}}{1} = -72$ | $\dfrac{-\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix}}{1} = 0$ | $\dfrac{-\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}}{1} = 0$ |
| $s^0$ | $\dfrac{-\begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix}}{-72} = 103$ | $\dfrac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$ | $\dfrac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$ |

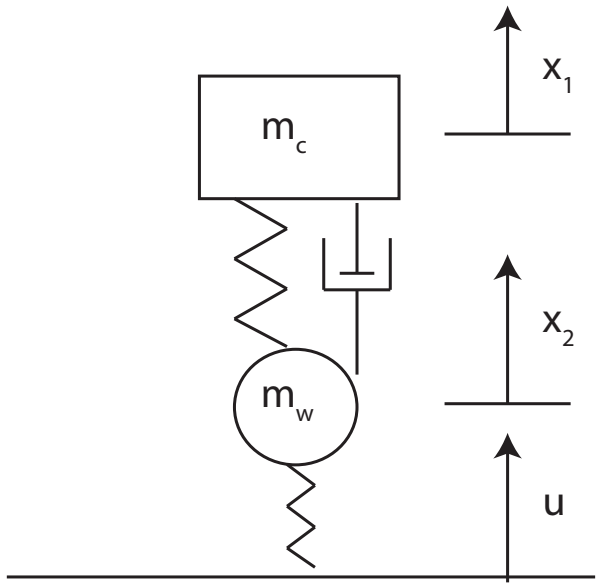
- We divide the second row by 10
- There are **two** sign changes: $1 \to -72$ and $-72 \to 103$
  - Two poles in the CRHP.

Feedback is **Destabilizing**!

# Another Numerical Example

Recall the suspension Problem with feedback:
**Closed Loop Transfer Function:**

$$\frac{k(s^2+s+1)}{s^4+2s^3+(3+k)s^2+(1+k)s+(1+k)}$$




**Question:** Can feedback destabilize the suspension system?

- Is it stable for any $k > 0$???

Lets start the Routh Table:

| | | | |
|---|---|---|---|
| $s^4$ | 1 | $3+k$ | $1+k$ |
| $s^3$ | 2 | $1+k$ | 0 |
| $s^2$ | $b_1$ | $b_2$ | $b_3$ |

We need to find the $b_i$.

# Another Numerical Example

Start by calculating the coefficients in the first row:

$$b_1 = -\frac{\det\begin{vmatrix} 1 & 3+k \\ 2 & 1+k \end{vmatrix}}{2} = \frac{1}{2}(5+k)$$

and

$$b_2 = -\frac{\det\begin{vmatrix} 1 & 1+k \\ 2 & 0 \end{vmatrix}}{2} = 1+k$$

which gives

| | | | |
|---|---|---|---|
| $s^4$ | 1 | $3+k$ | $1+k$ |
| $s^3$ | 2 | $1+k$ | 0 |
| $s^2$ | $\frac{1}{2}(5+k)$ | 1+k | 0 |
| $s$ | $c_1$ | 0 | 0 |

So far, so good.

- Now calculate the next row.

# Another Numerical Example

The coefficients for the next row are

$$c_1 = -\frac{\det \begin{vmatrix} 2 & 1+k \\ \frac{1}{2}(5+k) & 1+k \end{vmatrix}}{\frac{1}{2}(5+k)}$$
$$= \frac{k^2+2k+1}{5+k}$$

and $c_2 = 0$.

| | | | |
|---|---|---|---|
| $s^4$ | 1 | $3+k$ | $1+k$ |
| $s^3$ | 2 | $1+k$ | 0 |
| $s^2$ | $\frac{1}{2}(5+k)$ | $1+k$ | 0 |
| $s$ | $\frac{k^2+2k+1}{5+k}$ | 0 | 0 |
| 1 | $d_1$ | 0 | 0 |

Again, the first column is all positive for any $k > 0$

- Now calculate the final row.

# Another Numerical Example

There is only one non-zero coefficient in the last row.

$$d_1 = -\frac{\det \begin{vmatrix} \frac{1}{2}(5+k) & 1+k \\ \frac{k^2+2k+1}{5+k} & 0 \end{vmatrix}}{\frac{k^2+2k+1}{5+k}} = k+1$$

| | | | |
|---|---|---|---|
| $s^4$ | 1 | $3+k$ | $1+k$ |
| $s^3$ | 2 | $1+k$ | 0 |
| $s^2$ | $\frac{1}{2}(5+k)$ | 1+k | 0 |
| $s$ | $\frac{k^2+2k+1}{5+k}$ | 0 | 0 |
| 1 | $1+k$ | 0 | 0 |

**Conclusion:** No matter what $k > 0$ is, the first column is always positive.

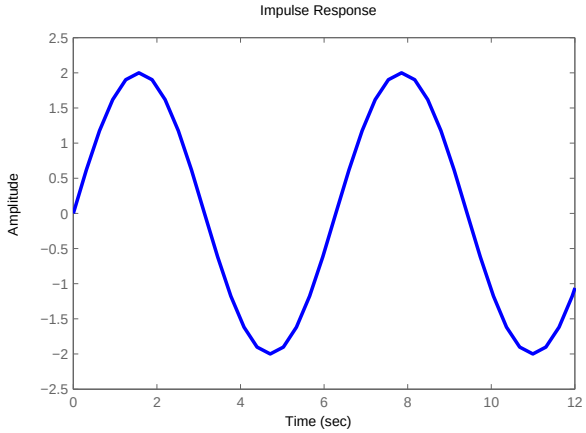
- No sign changes for any $k$.
- Stable for any $k$.
- We'll find out why later on.

Feedback **CANNOT** destabilize the suspension system.

# Stability of Quadratics

What about a simple second-order system?

$$\frac{1}{s^2 + bs + c}$$




We know the poles are at

$$p_{1,2} = -b \pm \sqrt{b^2 - 4c}$$

Calculate

$$-\frac{\det \begin{vmatrix} 1 & c \\ b & 0 \end{vmatrix}}{b} = -\frac{-bc}{b} = c$$

The Routh table is

| | | | |
|---|---|---|---|
| $s^2$ | $1$ | $c$ | $0$ |
| $s$ | $b$ | $0$ | $0$ |
| $1$ | $c$ | $0$ | $0$ |

Thus a quadratic is stable if and only if both coefficients are positive.

# Stability of 3rd order systems

Now consider a third order system:

$$\frac{1}{s^3 + as^2 + bs + c}$$

$$-\frac{\det\begin{vmatrix} 1 & b \\ a & c \end{vmatrix}}{a} = -\frac{c - ab}{a} = b - \frac{c}{a}$$

$$-\frac{\det\begin{vmatrix} a & c \\ b - \frac{c}{a} & 0 \end{vmatrix}}{b - \frac{c}{a}} = c$$

The Routh table is

| | | | |
|---|---|---|---|
| $s^3$ | $1$ | $b$ | $0$ |
| $s^2$ | $a$ | $c$ | $0$ |
| $s$ | $b - \frac{c}{a}$ | $0$ | $0$ |
| $1$ | $c$ | $0$ | $0$ |

So for 3rd order, stability is equivalent to:

- $a > 0$
- $c > 0$
- $b > \frac{c}{a}$

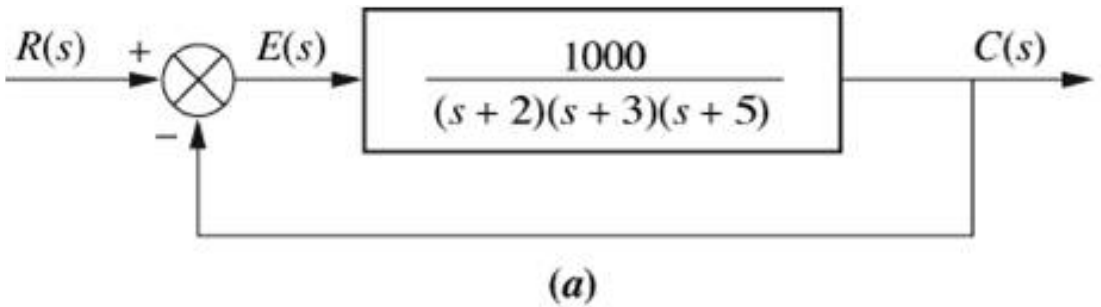
# Routh's Method

## Numerical Example, Revisited

Now lets look at the previous example to determine the maximum gain:
We have the stable transfer function

$$\hat{G}(s) = \frac{1}{(s+2)(s+3)(s+5)}$$

We close the loop with a gain of size $k$
**Controller:** $\hat{K}(s) = k$



(a)

The Closed-Loop Transfer Function is

$$\frac{k}{s^3 + 10s^2 + 31s + 30 + k}$$

But this is a third order system!

# Routh's Method

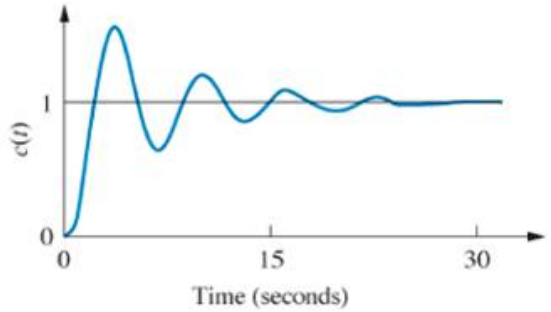
## Numerical Example, Revisited

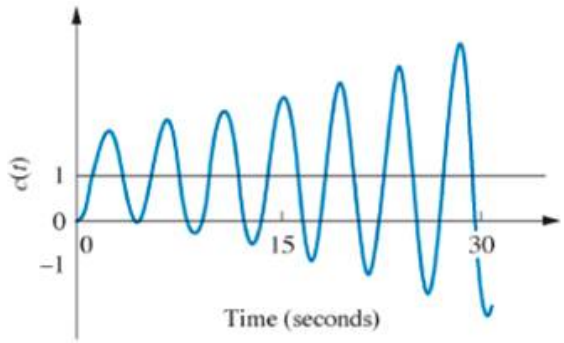
For the third-order system,

$$\frac{k}{s^3 + 10s^2 + 31s + 30 + k}$$

we require

- $a > 0$, which means $10 > 0$
- $c > 0$, which means $30 + k > 0$
- $b > \frac{c}{a}$




The last requirement implies $31 > \frac{k+30}{10}$ or

$$k < 310 - 30 = 280$$

So our gain is limited to $k < 280$

# Limited Special Cases

Consider the transfer function

$$\hat{G}(s) = \frac{1}{s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10}$$

The Routh Table begins:

| | | | |
|---|---|---|---|
| $s^5$ | 1 | 2 | 11 |
| $s^4$ | 2 | 4 | 10 |
| $s^3$ | 0 | 6 | 0 |

The next entry in the table will be

$$-\frac{\det \begin{vmatrix} 2 & 4 \\ 0 & 6 \end{vmatrix}}{0} = \frac{-12}{0}$$

Which is problematic.

**Note:** If there is a zero in the first column, the system is only marginally stable

- Small changes in the coefficients lead to instability.

# Limited Special Cases

The solution is to use $\epsilon$ instead of $0$ in the first column.

| | | | |
|---|---|---|---|
| $s^5$ | 1 | 2 | 11 |
| $s^4$ | 2 | 4 | 10 |
| $s^3$ | $\epsilon$ | 6 | 0 |

Now the next entry in the table will be

$$-\frac{\det\begin{vmatrix}2 & 4\\ \epsilon & 6\end{vmatrix}}{\epsilon} = \frac{-(12-4\epsilon)}{\epsilon}$$

Because $\epsilon$ is infinitely small, we let $12 - 4\epsilon = 12$.
Assume $\epsilon > 0$

- We have at least one sign change
- At least one unstable pole.

# Limited Special Cases

We can keep calculating if necessary.

| | | | |
|---|---|---|---|
| $s^5$ | 1 | 2 | 11 |
| $s^4$ | 2 | 4 | 10 |
| $s^3$ | 0 | 6 | 0 |
| $s^3$ | $\epsilon$ | 6 | 0 |
| $s^2$ | $\frac{4\epsilon - 12}{\epsilon}$ | $\frac{10\epsilon}{\epsilon}$ | 0 |
| $s^2$ | $\frac{-12}{\epsilon}$ | 10 | 0 |
| $s$ | $\frac{10\epsilon^2+72}{12}$ | 0 | 0 |
| $s$ | 6 | 0 | 0 |
| 1 | 10 | 0 | 0 |

So there are two sign changes

- Two unstable poles

# Limited Special Cases

Consider the transfer function

$$\hat{G}(s) = \frac{1}{s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56}$$

The Routh Table begins:

| | | | |
|---|---|---|---|
| $s^5$ | 1 | 6 | 8 |
| $s^4$ | 7 | 42 | 56 |
| $s^3$ | 0 | 0 | 0 |

The next entry in the table will be

$$-\frac{\det \begin{vmatrix} 7 & 42 \\ 0 & 0 \end{vmatrix}}{0} = \frac{0}{0}$$

Which is even more problematic - the whole row is zero.
We won't cover this case.

- However, it can be done - see book.

# Summary

What have we learned today?

The Routh-Hurwitz Stability Criterion:

- Determine whether a system is stable.
- An easy way to make sure feedback isn't destabilizing
- Construct the Routh Table

**Next Lecture: PID Control**